

CX 4220/CSE 6220 High Performance Computing
Midterm I Solutions

$$1. \begin{aligned} S(p) &= \frac{4n}{8\frac{n}{p} + 1000 \log p} \\ E(p) &= \frac{4n}{8n + 1000p \log p} \\ E(p) \geq \frac{1}{3} &\Rightarrow 1000p \log p \leq 4n \Rightarrow n \geq 250p \log p \end{aligned}$$

$$(a) \text{ For } n \gg p, S(p) \rightarrow \frac{4n}{8\frac{n}{p}} = \frac{p}{2}.$$

$$(b) 250 p \log p \leq 2,500,000 \Rightarrow p \log p \leq 10,000 \Rightarrow p \lesssim 1000.$$

$$(c) n > 250 \times 64 \times \log 64 = 250 \times 64 \times 6 = 96,000.$$

$$2. E(p) = \Theta\left(\frac{n}{n \log n + p \log p}\right)$$

$$(a) \text{ Maximum efficiency possible is } \Theta\left(\frac{n}{n \log n}\right) = \Theta\left(\frac{1}{\log n}\right).$$

(b) For achieving maximum possible efficiency, $p \log p = O(n \log n)$. Hence, $p \leq n \Rightarrow$ a maximum of n processors can be used.

3. Consider the following identity:

$$\begin{bmatrix} s_i & 1 \end{bmatrix} = \begin{bmatrix} s_{i-1} & 1 \end{bmatrix} \begin{bmatrix} x_i & 0 \\ y_i & 1 \end{bmatrix}$$

Therefore, compute parallel prefix on matrices

$$\begin{bmatrix} x_1 & 0 \\ y_1 & 1 \end{bmatrix}, \begin{bmatrix} x_2 & 0 \\ y_2 & 1 \end{bmatrix}, \begin{bmatrix} x_3 & 0 \\ y_3 & 1 \end{bmatrix}, \dots, \begin{bmatrix} x_{n-1} & 0 \\ y_{n-1} & 1 \end{bmatrix}$$

using matrix multiplication as the operator, and use the prefix matrix products to compute $s_1, \dots, s_{n-1}$. The run-time is the same as parallel prefix.

Computation time: $O\left(\frac{n}{p} + \log p\right)$

Communication time: $O((\tau + \mu) \log p)$

4. Define operator $\oplus$ such that

$$a \oplus b = \begin{cases} b, & \text{if } b \neq 0 \\ a, & \text{if } b = 0 \end{cases}$$

Essentially, $\oplus$ propagates value (a) to the right if the right element (b) is zero, and it is easy to show $\oplus$ is associative. Parallel prefix using $\oplus$ achieves *Array Broadcast*.

5. **for** $i = 0$ to $(\log p - 1)$ **do**
 exchange and balance elements with (rank XOR 2^i)
endfor

There are $\log p$ iterations during each of which at most n_{max} elements are exchanged. Hence,

Computation time = $O(n_{max} \log p)$

Communication time = $O((\tau + \mu n_{max}) \log p)$